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EEE 312 — Past Questions & Materials

by Renance

Course: EEE 312

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 3

Levels: 300

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Paper 1 of 3 — Digital Systems Fundamentals EEE 312

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES,

EFFURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY,

EXAMINATION: First Semester

SESSION: 2024/2025 0: COURSE CODE: FFI 312?

DATE: 21/05/2025 TIME ALLOWED: 3Hours

INSTRUCTIONS:

Answer five (5) questions in ALL : three (3) from Section A

necessary workings and include well-labelled diagrams

inscriptions on the question paper

and two (2) from Section B. Show

is where required No markings, notes, or

SECTION A

QUESTIONS ONE “

a) (1) FUPRE aims to develop an energy-efficient digital controller for advanced automation

systems.

Given that power consumption in digital

circuits is influenced by logic transitions and gate operations, how

can Boolean algebra

principles systematically be applied to:

i. Minimize switching activity

ii. Reduce gate count

iii. Enhance overall power efficiency in circuit design

iv. Provide a detailed explanation with relevant Boolean techniques and optimization Strategies.

(6 Marks)

(11) Derive the Boolean expression for the given circuit diagram:

A B C c

t

p-----L

y

\$---_1 a oe

a

(3 Marks)

b) Consider the truth table below: a

Literals Minterm - \Maxterm Sains iz ee

A Be ϕ xX) X2 : | __R i. J

ces — | A*Br+C so

A+B+C 0

are |

eee) A ch Sn i

A+B-+C 0

AtB-C 0

r - _

| A+B- ϕ fi

{| ATBrC me Saeed —_

Supp Wwe I 18 the Ulput, determine:

And fon and Maxterm expressions,

41) The VOR 1...: —

| R logic gate realization of the function

2 Marks)

(3 Marks)

(1) The Minterm

QUESTION TWO

a) (1) Mr. James K. say S that

) | os , : icts form

any Boolean function can be written in minimal sum-0! prod

as the sum of all of

. " » 1 7 . . p are J 1 > NS

the prime implicants of the function. Ben T. says that there are some functi

allo , a

Explain why | On does not involve all of the prime implicants.

-Xplain why James is rioht , | , 7

| > s4Mes IS right or provide a counter example demonstrating Ben's point.

whose minimal equ

(6 Marks)

m . vending machine system requires a logic circuit that detects when an exact amount (€8-. H10 In binary) is inserted. Design the circuit using logic gates. (3 Marks)

b) (i) How does the Transistor-Transistor Logic (TTL) family improve upon Diode-Transistor Logic (DTL)? (2 Marks)

(ii) Explain the Concept of Noise Margin and Its Variations Across TTL, CMOS, and ECL Logic Families (3 Marks)

QUESTION THREE

a) (1) With the aid of a well-labelled diagram, explain the structure and operational principles of a

three-state TTL (Transistor-Transistor Logic) gate. (6 Marks)

(ii) Compare the totem-pole output with open-collector and tri-state outputs, highlighting its advantages in terms of speed, output drive strength, and reduced power dissipation. (3 Marks)

b) (1) Given a CMOS gate with a specified output drive strength and an input capacitance per gate, derive an expression to determine its fan-out. (3 Marks)

(i) Explain the significance of Moore's Law in MOS/CMOS technology scaling. (2 Marks)

QUESTION FOUR

a) (i) Explain the historical evolution of integrated circuits (ICs) from SSI to VLSI.

What technological advancements enabled this progression? (6 Marks)

(ii) Analyze the influence of nanoelectronics and FinFET technology on the advancement of large-scale integration. (3 Marks)

b) (i) How does hardware-software co-design play a role in the development of VLSI-based systems?

(3 Marks)

(ii) Describe how multi-core processors and SoC (System-on-Chip) benefit from VLSI technology. (2 Marks)

ae.

SECTION R

QUESTION FIV]

dao

b) (1) design 4 bit ts register using NOR gales

(ii) What is a atch and design a gated latch using NAND gates

QUESTION SIX

2 a) (i) What are flipflops and why are they the essential building blocks of sequential circuits.

(ii) With the aid of a state table and diagram, explain the differences between SR flipflop and a JK flipflop.

b) (i)Determine the output of Q and /Q from the following signals assuming the flipflop is initially on RESE]

ee

» a oie

(a) () |

CLK Sat Or

(a) Di nae =

(ii) Distinguish between as\ nchronous and synchronous counter W vith the aid of a diagram.

(14 Marks)

QUESTION SEVE'

f

1) (1) Deduce the state table from the state diagram give below

xO 0)

11 0 1]

: Lx/] :

i)0} ~ dI

Oy (00.0 ())/] 00 []

v v

10 ~ ~ li

. 1x0 "

1/0

OI ix/]

O]/]

(ii) What are astable, bistable and monostable multivibrators and give 4 diagram and explanation for both monostable and astable multivibrators.

b) (i) A 555 timer is configured to run in a stable mode has the values $V_{ec} = 5.5V$, $R_1 = 2.2K\Omega$, $R_2 = 4.7K\Omega$, $C_1 = 0.022\mu f$ and $C_2 = 0.0\mu f$. Determine the output frequency and duty cycle.

< irate jus Dr itoul

(i:) Discuss the stages involved in drettal to analogue conversion and differentiate between analog to digital signals

(14 Marks)

| (oxt)4

Paper 2 of 3 — EEE 312 digital systems fundamentals 2025

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFERUN.
COLLEGE OF ENGINEERING AND TECHNOLOGY

EXAMINATION: First Semester
SESSION: 2024/2025
DATE: 21/05/2025

COURSE TITLE: Digital Systems Fundamentals
COURSE CODE: EEE 312
TIME ALLOWED: 3 Hours

INSTRUCTIONS:
Answer five (5) questions in ALL: three (3) from Section A and two (2) from Section B. Show all necessary workings and include well-labelled diagrams where required. No markings, notes, or inscriptions on the question paper.

SECTION A

QUESTIONS ONE

a) (i) FUPRE aims to develop an energy-efficient digital controller for advanced automation systems. Given that power consumption in digital circuits is influenced by logic transitions and gate operations, how can Boolean algebra principles systematically be applied to:

- Minimize switching activity
- Reduce gate count
- Enhance overall power efficiency in circuit design
- Provide a detailed explanation with relevant Boolean techniques and optimization strategies.

(6 Marks)

(ii) Derive the Boolean expression for the given circuit diagram:

(3 Marks)

b) Consider the truth table below:

Literals			Minterm	Maxterm	
A	B	C	X_1	X_2	R
0	0	0	ABC	$A+B+C$	0
0	0	1	ABC	$A+B+C$	0
0	1	0	ABC	$A+B+C$	1
0	1	1	ABC	$A+B+C$	1
1	0	0	ABC	$A+B+C$	0
1	0	1	ABC	$A+B+C$	0
1	1	0	ABC	$A+B-C$	1
1	1	1	ABC	$A+B+C$	1

Suppose R is the output, determine:

- (i) The Minterm and Maxterm expressions. (2 Marks)
- (ii) The AND/OR logic gate realization of the function. (3 Marks)

QUESTION TWO

- a) (i) Mr. James K. says that any Boolean function can be written in minimal sum-of-products form as the sum of all of the prime implicants of the function. Ben T. says that there are some functions whose minimal equation does not involve all of the prime implicants.

Explain why James is right or provide a counter example demonstrating Ben's point. (6 Marks)

- (ii) A vending machine system requires a logic circuit that detects when an exact amount (e.g., 110 in binary) is inserted. Design the circuit using logic gates. (3 Marks)

- b) (i) How does the Transistor-Transistor Logic (TTL) family improve upon Diode-Transistor Logic (DTL)? (2 Marks)
- (ii) Explain the Concept of Noise Margin and Its Variations Across TTL, CMOS, and ECL Logic Families (3 Marks)

QUESTION THREE

- a) (i) With the aid of a well-labelled diagram, explain the structure and operational principles of a three-state TTL (Transistor-Transistor Logic) gate. (6 Marks)

- (ii) Compare the totem-pole output with open-collector and tri-state outputs, highlighting its advantages in terms of speed, output drive strength, and reduced power dissipation. (3 Marks)

- b) (i) Given a CMOS gate with a specified output drive strength and an input capacitance per gate, derive an expression to determine its fan-out. (3 Marks)

- (ii) Explain the significance of Moore's Law in MOS/CMOS technology scaling. (2 Marks)

QUESTION FOUR

- a) (i) Explain the historical evolution of integrated circuits (ICs) from SSI to VLSI. What technological advancements enabled this progression? (6 Marks)

- (ii) Analyze the influence of nanoelectronics and FinFET technology on the advancement of large-scale integration. (3 Marks)

- b) (i) How does hardware-software co-design play a role in the development of VLSI-based systems? (3 Marks)

- (ii) Describe how multi-core processors and SoC (System-on-Chip) benefit from VLSI technology. (2 Marks)

SECTION B

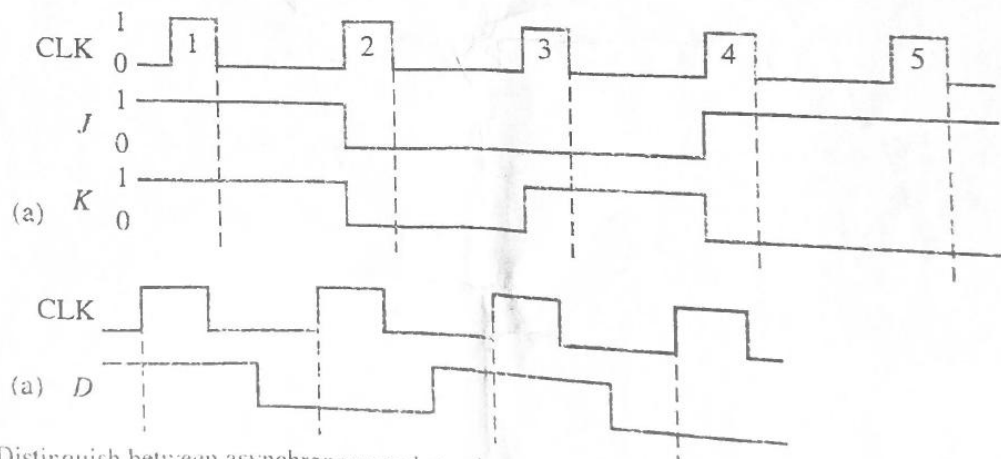
QUESTION FIVE

- a) (i) Discuss combinational and sequential circuits and how they differ.
 (ii) Differentiate between synchronous and asynchronous with the aid of a block diagram
- b) (i) design 4 bits register using NOR gates.
 (ii) What is a latch and design a gated latch using NAND gates.

(14 Marks)

QUESTION SIX

- 2 a) (i) What are flipflops and why are they the essential building blocks of sequential circuits.
 (ii) With the aid of a state table and diagram, explain the differences between SR flipflop and a JK flipflop.
- b) (i) Determine the output of Q and /Q from the following signals assuming the flipflop is initially on RESET.



- (ii) Distinguish between asynchronous and synchronous counter with the aid of a diagram.

(14 Marks)

Paper 3 of 3 — EEE 312 Past Question and Answers

Level 300 • First Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING**

**EXAMINATION: First Semester
SESSION: 2020/2021**



**COURSE CODE: EEE 312
COURSE TITLE: Digital Systems
Fundamentals
ALLOWED: 3Hrs**

DATE: 18/11/21

**INSTRUCTION: ANSWER FIVE QUESTIONS WITH AT LEAST TWO QUESTIONS
FROM EACH SECTION**

SECTION A

Q1. (a) Using the Resistor Transistor Logic (RTL) technology, design a three input OR and NAND gate. For each of the design, explain using the circuit drawn what the output will be when the input is 011 and 111. 4 Marks

(b) Draw using logic gates the boolean expression Equation 1.1, simplify the boolean expression using the laws of Boolean algebra and Demorgan's theorem, draw the circuit of the simplified expression using logic gates. 10 marks

$$Y = (\overline{A}BC + A\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C}) + D \quad 1.1$$

Q2. (a) Design a logic circuit for a simple Automobile door and seat belt buzzer system. Assume that sensors are available to provide digital signals representing the door and seat belt states. Signal A is from the door, where HIGH implies the door is closed; signal B is from the seat belt, where HIGH implies the seat belt is fastened; and signal C indicates whether the ignition switch is turned ON. Your circuit should output a signal X that can be used to turn a buzzer ON or OFF, where HIGH implies ON. The buzzer should be ON if the ignition is ON and if the door is open or the seat belt is not fastened. Draw the truth table and use K-map minimization to write the Boolean expression for the system. Draw the logic circuit to implement the Boolean expression. 6 marks

(b) Find the Boolean expression for the output P of the circuit shown in figure 2.1. Draw the truth table to show the outputs at M, N, O and P of each of the gate assuming the inputs are 001, 010, 011, 100, 101, 110 and 111 8Marks

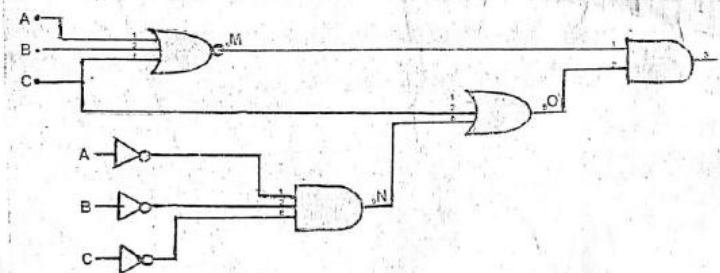


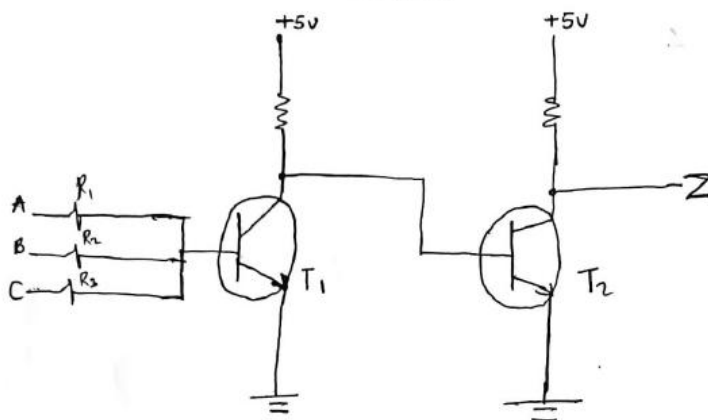
Figure 2.1

3. Design a synchronous counter that has the following repeated binary sequence: 0, 1, 2, 3, 4, 5, 6, and 7. Draw the counter using R-S Flip flop. 14 marks
4. (a) Design a synchronous counter that has the following repeated binary sequence: 0, 2, 3, 5, 6. All unused state should be taken as don't care during the next state. Draw the counter using Delay Flip flop. 11 marks
- (b) Implement the following expression using Karnaugh Map minimization and Draw the Logic circuit. $F(A,B,C) = \Sigma(1, 3, 4, 5, 6, 7)$ 3 marks

EEE312 [2020/2021] SOLUTION

Question 1

a) RTL OR Gate <Three input>



✓ When any of the three input is at logic 1, the first transistor T_1 is biased, making current to flow into the ground thereby making its output which is fed into the input of the second resistor low at logic '0'. This makes the second transistor not to bias (i.e. current will not flow) making the output Z of the transistor T_2 to be High at logic 1.

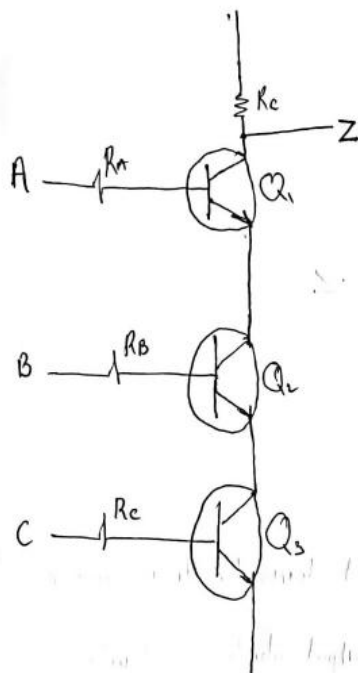
Logic output

A	B	C	Z
0	1	1	1

✓ When the three inputs are high at logic 1, the current flows from the source to the ground making the output of the first transistor which is fed into the input of the other be low at '0' (i.e. the transistor is biased). This makes the second transistor ^{not} to be biased making output Z to be HIGH at '1'.

A	B	C	Z
1	1	1	1

RTL NAND Gate (three inputs)



✓ When the input is 011, this in this case the transistors ^{Q_1 is not} bases are LOW at logic '0' cutting the current from flowing to the ground thereby making Z output to be HIGH at 1

✓ When the input is 111, the three transistors are bases making the current to flow from the source to the ground thereby having '0' at the output Z as "Low"

$$b \quad Y = \overline{(A\bar{B}C + A\bar{B}\bar{C}\bar{D} + \bar{A}BCD + \bar{A}\bar{B}CD)} + D$$

$$\text{Letting } A\bar{B}C + A\bar{B}\bar{C}\bar{D} + \bar{A}BCD + \bar{A}\bar{B}CD = P$$

$$Y = \overline{P} + D$$

By applying demorgan's theorem

$$Y = \overline{\overline{P} + D}$$

$$Y = \overline{P} \cdot \bar{D}$$

$$Y = \bar{P} \bar{D}$$

$$Y = \overline{\overline{P} \bar{D}}$$

$$Y = P \bar{D}$$

Replace P;

$$Y = (A\bar{B}C + A\bar{B}\bar{C}\bar{D} + \bar{A}BCD + \bar{A}\bar{B}CD) \bar{D}$$

$$Y = A\bar{B}C(\bar{D}) + A\bar{B}\bar{C}(\bar{D} \cdot \bar{D}) + \bar{A}BC(\bar{D} \cdot \bar{D}) + \bar{A}\bar{B}C(\bar{D} \cdot \bar{D})$$

$$\text{Where } D \cdot \bar{D} = 0$$

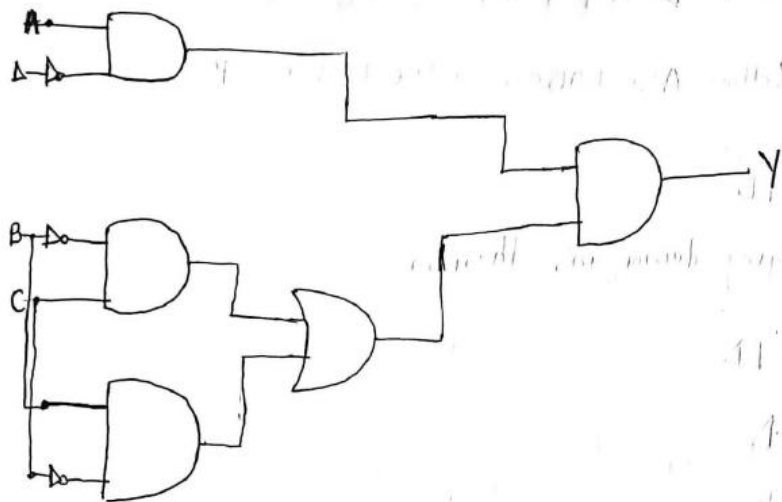
$$\bar{D} \cdot \bar{D} = \bar{D}$$

$$D \cdot D = D$$

$$Y = A\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D}$$

$$Y = A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D}$$

$$Y = A\bar{D}(\bar{B}C + \bar{B}\bar{C})$$



2a A; High 1 = close
Low 0 = open

B; High 1 = fastened
Low 0 = not fastened

C; High 1 = ON
Low 0 = off

Condition for X to be ON; If the ignition is ON and if the door is open ~~and~~ or the seat belt is not fastened.

Interpretation $\frac{1}{2}$ Truth Table

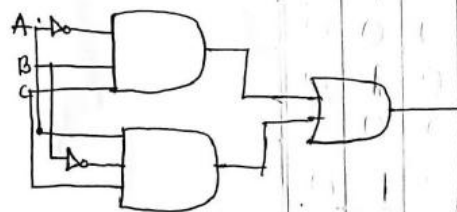
A	B	C	X
0	1	1	1
1	0	1	1

K-map

AB \ C	00	01	11	10
0				
1		1		1

Expression

$$X = \bar{A}BC + A\bar{B}C$$



b) $P = M * O$
 $M = (A + B + C)$
 $O = C$

2 (b) The boolean express is as follows

$$P = MO$$

$$M = (A + B + C)$$

$$O = C + N$$

$$N = \overline{A} \overline{B} \overline{C}$$

$$\therefore O = C + (\overline{A} \overline{B} \overline{C})$$

$$P = (A + B + C) [C + \overline{A} \overline{B} \overline{C}]$$

Truth table

	A	B	C	M	$\overline{A}$	$\overline{B}$	$\overline{C}$	N	O	P
1	0	0	1	0	1	1	0	0	1	0
2	0	1	0	0	1	0	1	0	0	0
3	0	1	1	0	1	0	0	0	1	0
4	1	0	0	0	0	1	1	0	0	0
5	1	0	1	0	0	1	0	0	1	0
6	1	1	0	0	0	0	1	0	0	0
7	1	1	1	0	0	0	0	0	1	0

③

Present state			Next state			Flip flops					
Q_c	Q_b	Q_a	Q_c	Q_b	Q_a	R_c	S_c	R_b	S_b	R_a	S_a
0	0	0	0	0	1	X	0	X	0	0	1
0	0	1	0	1	0	X	0	0	1	1	0
0	1	0	0	1	1	X	0	0	X	0	1
0	1	1	1	0	0	0	1	1	0	1	0
1	0	0	1	0	1	0	X	X	0	0	1
1	0	1	1	1	0	0	X	0	1	1	0
1	1	0	1	1	1	0	X	0	X	0	1
1	1	1	0	0	0	1	0	1	0	1	0

P_s	N_s	R	S
0	0	X	0
0	1	0	1
1	0	1	0
1	1	0	X

R_c input kmap

$Q_c Q_b$	00	01	11	10
Q_a 0	X	X		
Q_a 1	X		1	

$$R_c = Q_c Q_b Q_a$$

$Q_c Q_b$	00	01	11	10
Q_a 0			X	X
Q_a 1		1		X

$$S_c = \bar{Q}_c Q_b Q_a$$

R_b k map

$Q_c Q_b$	00	01	11	10
Q_a 0	X			X
Q_a 1		1	1	

$$R_b = Q_b Q_a$$

$Q_c Q_b$	00	01	11	10
Q_a 0		X	X	
Q_a 1	1			1

$$S_b = \bar{Q}_b Q_a$$

R_A kmap

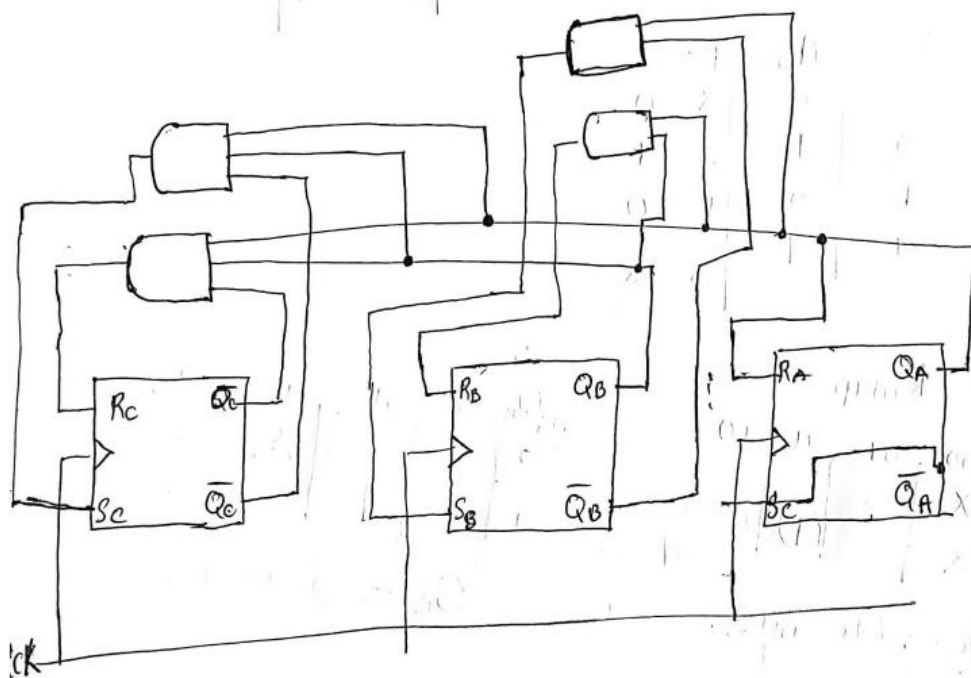
Q_A	Q_B	00	01	11	10
0					
1		1	1	1	1

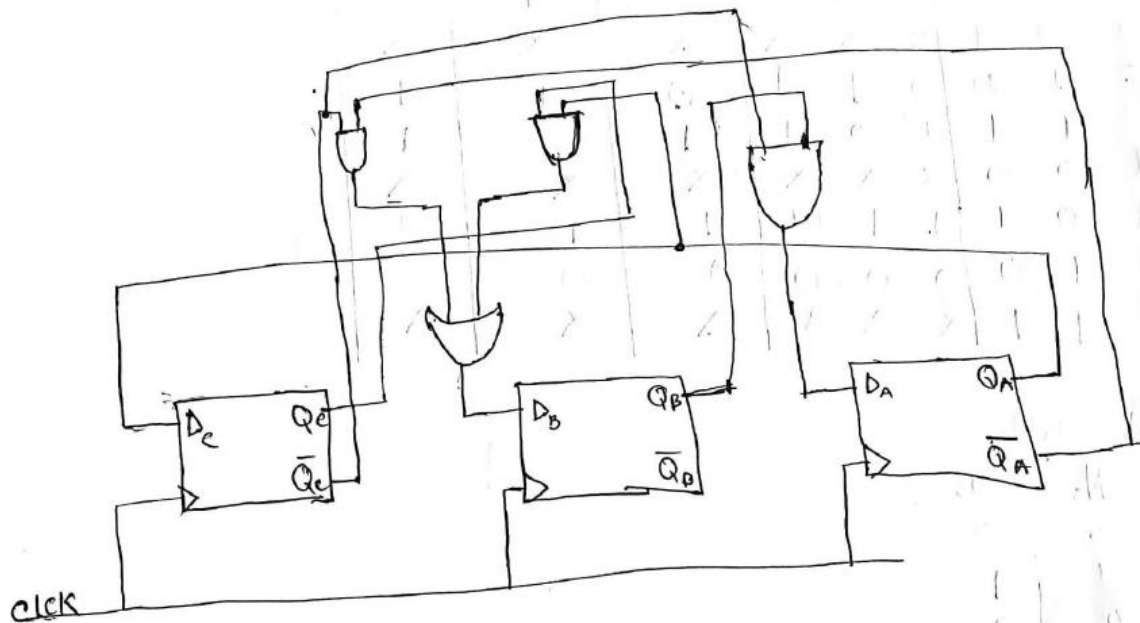
$$R_A = Q_A$$

S_A

Q_A	Q_B	00	01	11	10
0		1	1	1	1
1					

$$S_A = \bar{Q}_A$$





Ha

	Present State			Next State			Flip' Flop		
	Q_c	Q_B	Q_A	Q_c	Q_B	Q_A	D_c	D_B	D_A
0	0	0	0	0	1	0	0	1	0
1	0	0	1	X	X	X	X	X	X
2	0	1	0	0	1	1	0	1	1
3	0	1	1	1	0	1	1	0	1
4	1	0	0	X	X	X	X	X	X
5	1	0	1	1	1	0	1	1	0
6	1	1	0	0	0	0	0	0	0
7	1	1	1	X	X	X	X	X	X

Excitation Table

P_s	N_s	D
0	0	0
0	1	1
1	0	0
1	1	1

 D_c K map

$Q_c \backslash Q_A$	00	01	11	10
0				X
1	X	1	X	1

$$D_c = Q_A$$

 D_B K-map

$Q_c \backslash Q_A$	00	01	11	10
0	1	1		X
1	X		X	1

$$D_B = \bar{Q}_c \bar{Q}_A + Q_c Q_A$$

 D_A Kmap

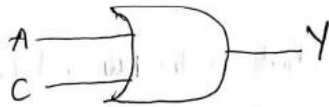
$Q_c \backslash Q_B$	00	01	11	10
0		1		X
1	X	1	X	

$$D_A = \bar{Q}_c Q_B$$

4b Using the k-map minimization

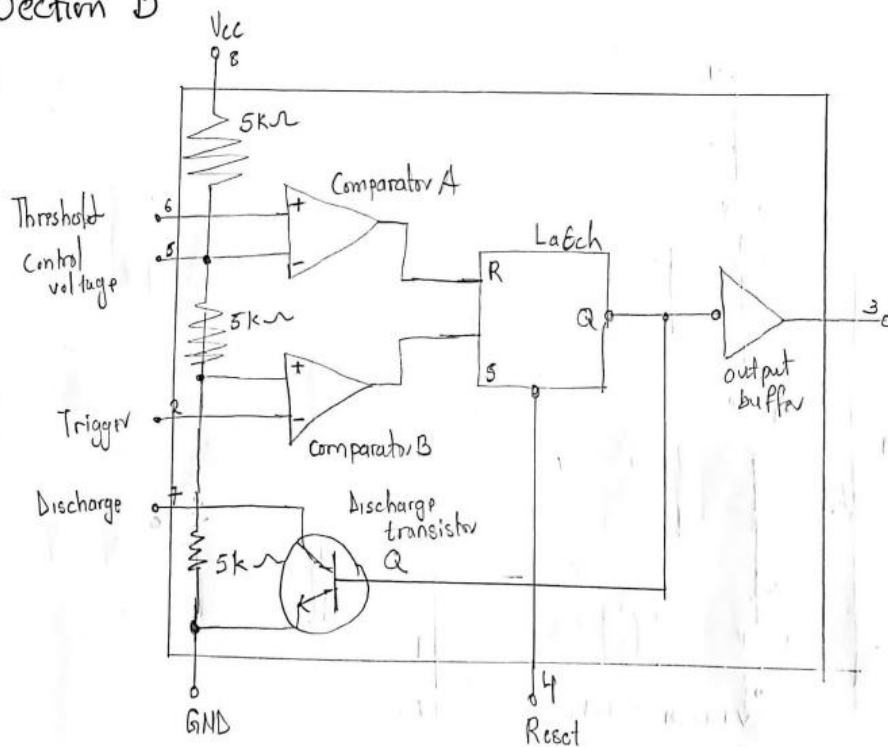
	AB	00	01	10	11
C	0			1	1
	1	1	1	1	1

$$Y = C + A = A + C$$



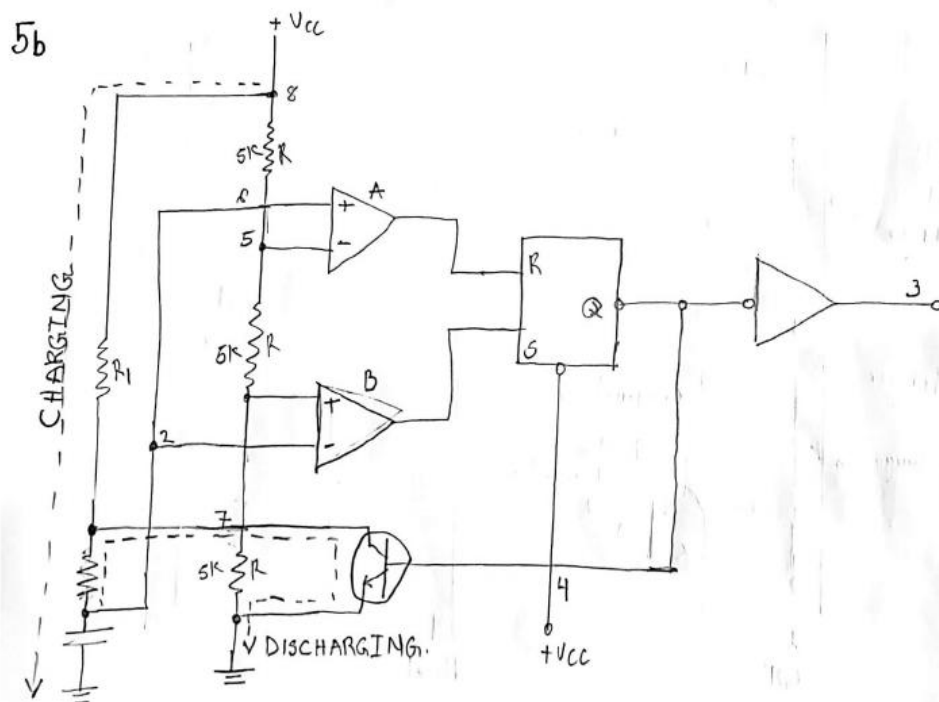
Section B

5a



The Internal functional diagram of a 555 timer

- * When the normally trigger input momentarily goes below $\frac{1}{3}V_{CC}$, the output of comparator B switches from LOW to HIGH making the RS latch output in pin 3 to be HIGH causing the transistor to off.
- * This output remain high until the low threshold input is about $\frac{2}{3}V_{CC}$ and it causes the output of Comparator A to switch from LOW to HIGH.
- * The reset pin 4 causes the output to be LOW and turning the transistor ON.
- * The trigger and threshold are controlled by external components connected to produce either monostable or astable action.



Operation of a 555 timer as an astable multivibrator

c) $R_1 = 2.4 \text{ k}\Omega$

$R_2 = 5 \text{ k}\Omega$

$C_1 = 0.01 \mu\text{F}$

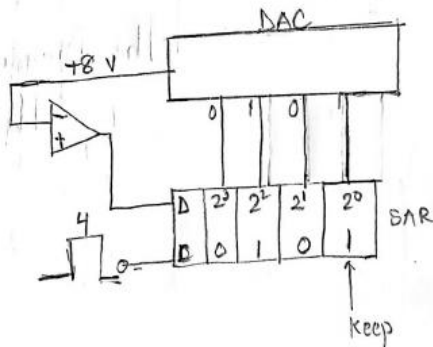
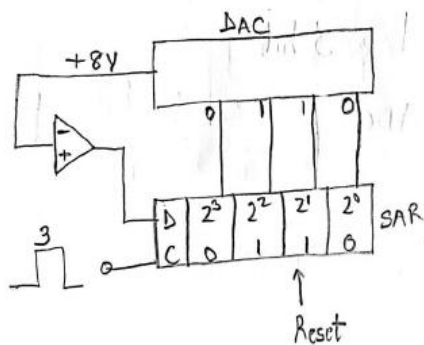
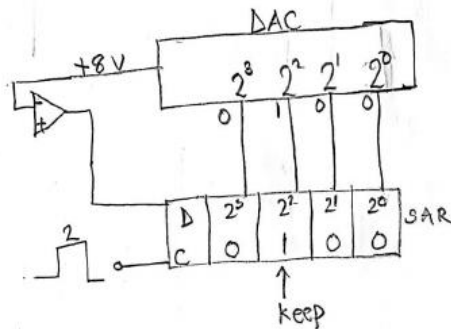
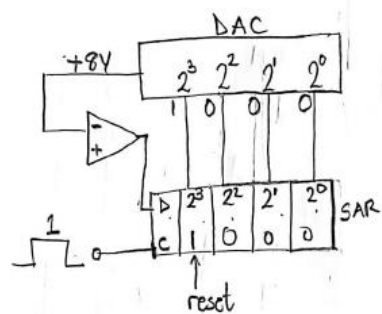
$$f = \frac{1.44}{(R_1 + 2R_2)C_1} = \frac{1.44}{(2.4 \times 10^3 + 2(5 \times 10^3))0.01 \times 10^{-6}}$$

$$= \frac{1.44}{(2.4 + 10) \times 0.01 \times 10^{-6} \times 10^3}$$

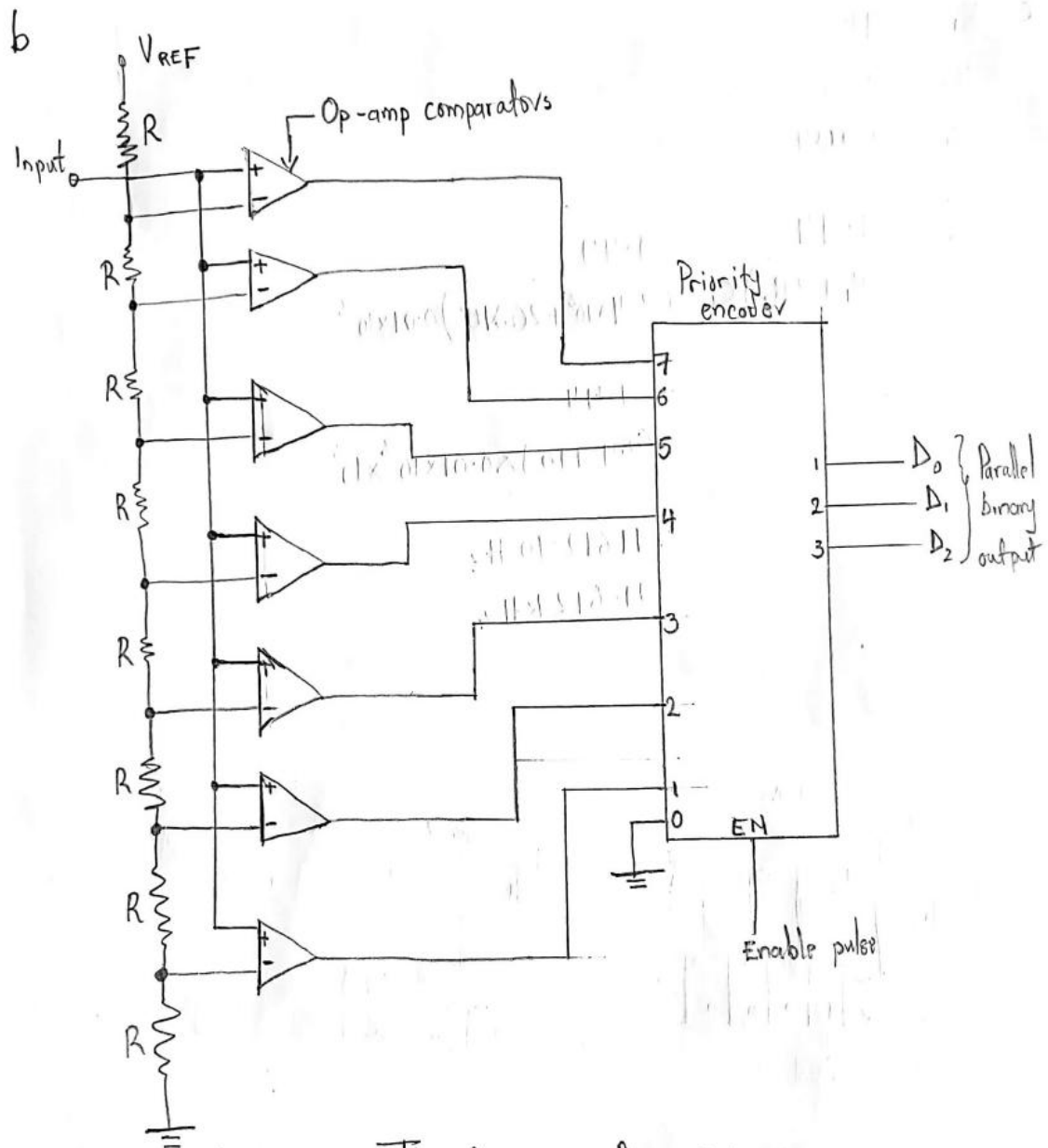
$$= 11612.90 \text{ Hz}$$

$$= 11.612 \text{ kHz}$$

6

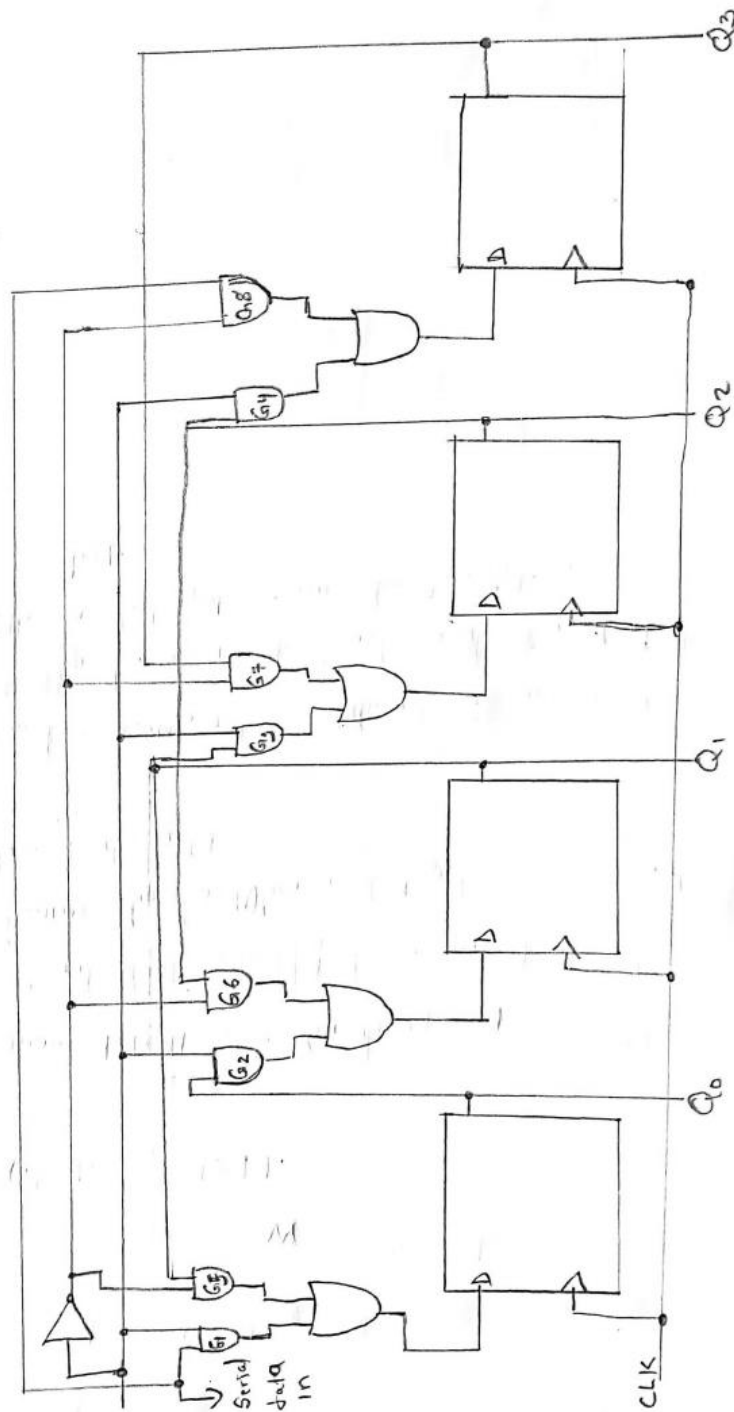


Successive Approximation ADC of 5.1V to a digital signal

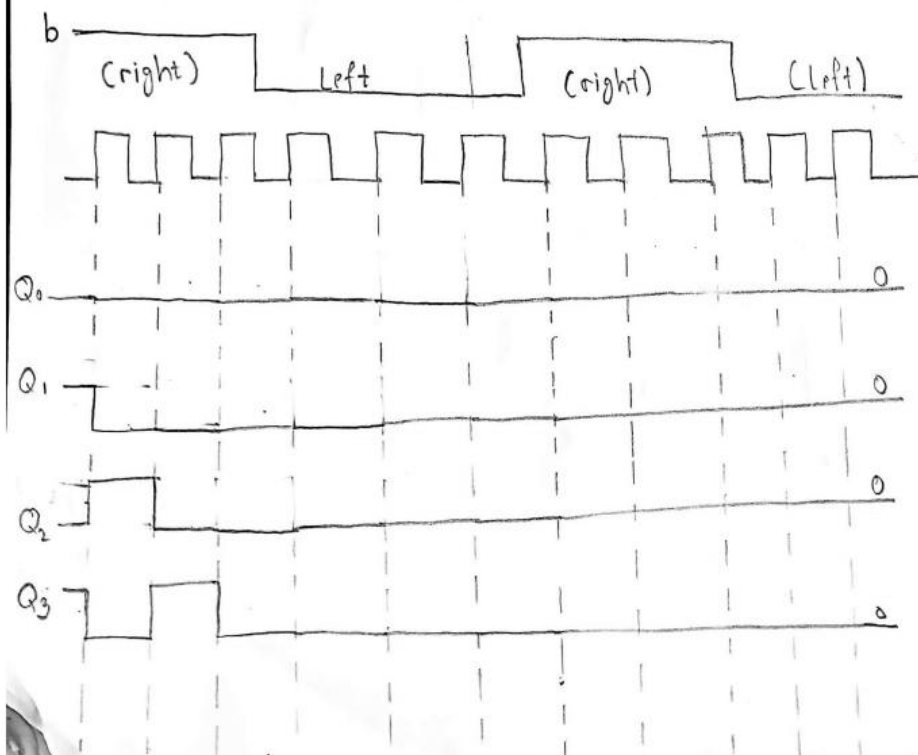


The Diagram of a 3-bit
flash ADC

7 a. 4-bit bidirectional shift register



- ✓ A HIGH on the control input allows data bits inside the register to be shifted to right and a LOW enables data bits inside the register to be shifted to ~~left~~ LEFT.
- ✓ When the control input HIGH, gates G_1 through G_4 are enabled, and the state of the Q output of each flip-flop is passed through to the Δ input of the following flip-flop. When a clock pulse occurs, the data bits are shifted one place to the right.
- ✓ When the control input is LOW, gates G_5 through G_8 are enabled and the output of each flip-flop is passed through to the Δ input of the preceding flip-flop. When a clock pulse occurs, the data bits are then shifted one place to the left.



The bidirectional shift register table of Operation is given below

Clock	Q ₀	Q ₁	Q ₂	Q ₃
0	0	1	0	1
1	0	0	1	0
2	0	0	0	1
3	0	0	0	0
4	0	0	0	0
5	0	0	0	0
6	0	0	0	0
7	0	0	0	0
8	0	0	0	0
9	0	0	0	0
10	0	0	0	0

↘ RIGHT

↙ LEFT

C Initial state = 0101001111

CLK	Q ₀	Q ₁	Q ₂	Q ₃	Q ₄	Q ₅	Q ₆	Q ₇	Q ₈	Q ₉
0	0	1	0	1	0	0	1	1	1	1
1	1	0	1	0	1	0	0	1	1	1
2	1	1	0	1	0	1	0	0	1	1
3	1	1	1	0	1	0	1	0	0	1
4	1	1	1	1	0	1	0	1	0	0
5	0	1	1	1	1	0	1	0	1	0
6	0	0	1	1	1	1	0	1	0	1
7	1	0	0	1	1	1	1	0	1	0
8	0	1	0	0	1	1	1	1	0	1
9	1	0	1	0	0	1	1	1	1	0

Note; i) In ring counter, the output at every clock pulse is directly fed back into the input by shifting the previous input and replacing it.

ii) At the last clock pulse your initial state must be replicated.